

Grade 7 Accelerated
Unit 1
Lesson 3 Homework

Name _____
Date _____
Period _____

Match the algebraic statement to its numeric example.

1.

Property
Symmetric Property of Equality
Addition Property of Equality
Subtraction Property of Equality
Multiplication Property of Equality
Division Property of Equality

Numeric Example
$5 - 4 = 3 + 2 - 4$
$5 \div 4 = (3 + 2) \div 4$
$5 \times 4 = (3 + 2) \times 4$
$5 = 3 + 2$
$5 + 4 = 3 + 2 + 4$

Consider the equation $2 + 3n = 11$ to answer questions 2-5.

2. Complete the statement to describe the equation.

The variable, n , is first being _____ by 3, and then 2 is being _____ to this product. The value of the expression, $2 + 3n$ is _____ to 11.

3. Describe the operation that could be performed to isolate $3n$ on the left side of the equation.
4. Describe the operation that could be performed to isolate the n on the left side of the equation.
5. Solve $2x - 5 = 11$ by applying properties of equality. Explain which properties were used to solve the equation.

6. Solve $\frac{n}{12} - 3 = 7$ by applying properties of equality. Explain which properties were used to solve the equation.

Use the equation $6y + 5 = 29$ to complete the following.

7. Apply properties of equality to solve the equation. Use substitution to verify the solution.

Use the equation $9a + 7 = 88$ to complete the following.

8. Apply properties of equality to solve the equation. Use substitution to verify the solution.

Use the equation $\frac{y}{4} - 8 = 4$ to complete the following.

9. Apply properties of equality to solve the equation. Use substitution to verify the solution.

Use the equation $-8 + \frac{b}{3} = 10$ to complete the following.

10. Apply properties of equality to solve the equation. Use substitution to verify the solution.